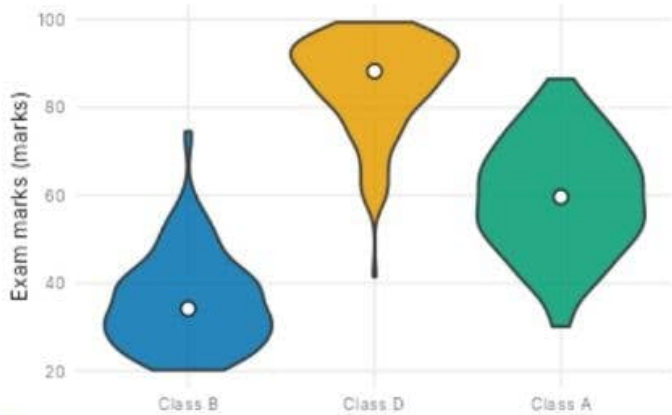


R.No: 10 Reg. No: 192424055 Name: DEEKSHITA U

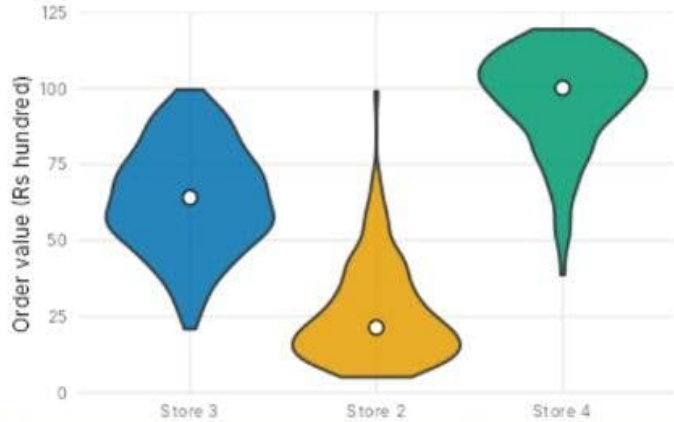
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Exam marks (marks)



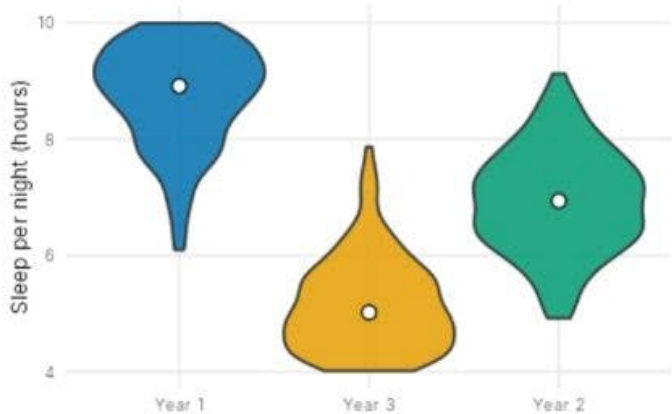
Violin	Skew (circle one)	Mean vs Median (circle)
Class B	Right / Left / Symm.	Mean > / < / ~ Median
Class D	Right / Left / Symm.	Mean > / < / ~ Median
Class A	Right / Left / Symm.	Mean > / < / ~ Median

Chart 2: Order value (Rs hundred)



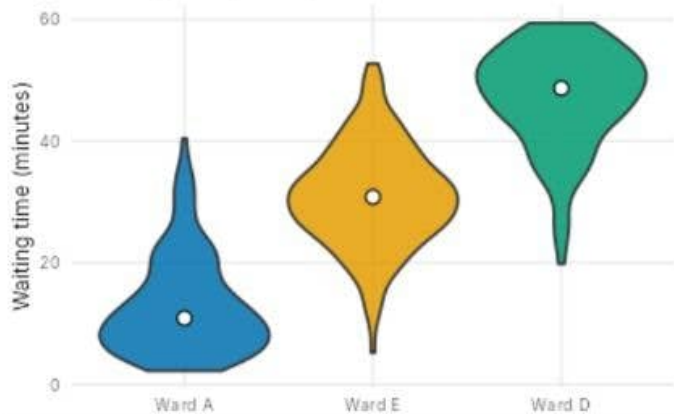
Violin	Skew (circle one)	Mean vs Median (circle)
Store 3	Right / Left / Symm.	Mean > / < / ~ Median
Store 2	Right / Left / Symm.	Mean > / < / ~ Median
Store 4	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Sleep per night (hours)



Violin	Skew (circle one)	Mean vs Median (circle)
Year 1	Right / Left / Symm.	Mean > / < / ~ Median
Year 3	Right / Left / Symm.	Mean > / < / ~ Median
Year 2	Right / Left / Symm.	Mean > / < / ~ Median

Chart 4: Waiting time (minutes)



Violin	Skew (circle one)	Mean vs Median (circle)
Ward A	Right / Left / Symm.	Mean > / < / ~ Median
Ward E	Right / Left / Symm.	Mean > / < / ~ Median
Ward D	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

B1. In Chart 1, violin "Class B" is not symmetric. Towards which side is its tail, and which is the better measure of a typical exam marks here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

10/8/2026

Roll no: 10

Deekshitha.U

ACTIVITY - A

VIOLIN SKEWNESS

PART - A

Chart 1: Exam Marks:

Class B: Most marks are around 25-45, with a long tail extending towards higher marks.

Right skew.

Class D: Most marks are around 80-95, with a longer tail extending towards lower marks.

Left skew.

Class A: Distribution is reasonably balanced around the centre.

Approximately skewed

Chart 2: Order Value.

Store 3: Main concentration is around 50-70, with a tail toward higher values.

Right skew.

Store 2: Most values are low with a long upper tail.

Right skew

Store 4: Most values are high, with a long lower tail.

Left skew.

Chart 3: Sleep per night:

Year 1: Most students sleep around 9-10 hrs with a tail towards lower values.
Left skew.

Year 2: Most values are around 4-5 hrs, with a tail higher values.

Right skew.

Year 3: Distribution is fairly balanced around 7 hrs.

Approximately symmetric.

Chart 4: Waiting Time:

Ward A: Most waiting times are low, with a long tail toward higher values.

Right skew.

Ward E: Values are reasonably balanced around the centre.

Approximately symmetric.

Ward D: Most values are high, with a tail toward lower values.

Left skew.

PART - B.

B1:

Class B is right-skewed, with a tail toward higher marks. The median is a better measure of a typical exam mark because the mean is pulled upward by the higher values.

B2:

```
ggplot(df, aes(x=group, y=value)) +  
  geom_violin() +  
  stat_summary(fun = median, geom = "point",  
               shape = 21, fill = "white")
```